



PASCHAL IHENACHO

AI-INTEGRATED FULL-STACK SOFTWARE ENGINEER

CONTACT

- 📞 832-641-2011
- ✉️ ihenachopaschal@yahoo.com
- 🏠 Houston, TX, USA
- 🌐 <https://www.linkedin.com/in/paschal-ihenacho/>

EDUCATION

Boot Camps

Full Stack Web Development,
University of Texas Austin
2020

Bachelor's Degree

Computer Science, University
of Houston-Downtown
2016

Associate's degree

Computer Science, Houston
Community College
2013

SKILLS

- JavaScript, TypeScript, Python, Node.js, NestJS, React, Angular, VueJS, Next.js, SQL
- OpenAI API, LangChain, RAG pipelines, Pinecone, AI-driven Automation Debugging
- Playwright, Jest, API Testing, TDD, Self-Healing Test Frameworks
- Azure DevOps, GitHub Actions, AWS Lambda, Docker, Vercel, CI/CD pipelines

PROFESSIONAL OVERVIEW

Full Stack Software Engineer with **8+ years of experience** building scalable web applications and **AI-powered software solutions**. Specialized in **TypeScript/JavaScript, Node.js, NestJS, React, and Next.js**, with hands-on experience integrating **LLMs using LangChain, Claude/OpenAI APIs, and vector databases (Pinecone)** to develop intelligent features such as **in-app AI assistants that guide users through ERP workflows using natural language queries and step-by-step instructions**. Experienced in designing **RESTful APIs, microservices architectures, and RAG pipelines**, with strong expertise in **PostgreSQL, TypeORM, CI/CD (Azure DevOps, GitHub Actions), and cloud deployments on AWS, Azure, and Vercel**.

WORK EXPERIENCE

Full Stack Developer

Enertia Software, Houston, TX

2023 – Present

- Designed and developed scalable web applications using **React** for dynamic front-end interfaces and **Node.js (ExpressJS, NestJS)** for robust backend services.
- Built and optimized **RESTful APIs** to support high-performance data exchange across distributed systems, utilizing **TypeORM** for data modeling and database interaction.
- Migrated legacy **SOAP-based APIs to modern REST services**, improving system interoperability, maintainability, and response performance.
- Configured API infrastructure including **structured response bodies, standardized error handling, and HTTP status code conventions** aligned with company engineering standards.
- **Integrated AI-powered application support using LLMs (Claude/OpenAI) and LangChain**, enabling end users to interact with the ERP platform through natural language queries.
- Developed an **AI assistant embedded within the ERP web application** that allows users to ask functional questions about system workflows and receive **step-by-step guidance on how to perform tasks within the application**.
- Implemented **Retrieval-Augmented Generation (RAG) pipelines using LangChain**, allowing the AI assistant to retrieve documentation, knowledge-base content, and internal application guides to generate accurate responses.
- Designed prompt orchestration and context management strategies to improve **response relevance, accuracy, and usability for ERP end users**.
- Built backend services that **connect ERP metadata, documentation, and user workflows to LLM-powered query systems**, enabling intelligent search and guided navigation through the platform.
- Optimized **WebSocket communication performance** by reducing payload sizes and implementing message throttling for real-time features.
- Secured WebSocket connections using **authentication tokens and role-based access control**, ensuring secure real-time data communication.
- Implemented **authentication and authorization guards in Node.js (NestJS Guards)** to enforce role-based access, token validation, and request filtering across API endpoints.

- Microservices Architecture, Fault-Tolerant Design, Scalable Web Solutions

PROFESSIONAL DEVELOPMENT

Certifications

- “Generative AI” – LinkedIn 2025

Additional Training

- “Python Essential Training” – LinkedIn 2025

Personal Project (EasyShop)

EasyShop is a production-grade React/NextJS-NestJS e-commerce platform built with TypeORM, MySQL, and NgRx, designed for scalability, maintainability, and modern shopping experiences across devices.

Key Highlights:

- Component-driven architecture for scalability and reuse
- Reactive, observable-based state flow
- Secure authentication with JWT and OAuth 2.0
- Seamless CI/CD automation
- Clean code and domain separation for maintainability

Tech Stacks:

AWS S3, NodeJS, NestJS, TypeORM, MySQL, RESTful API, JWT & OAuth 2.0, Typescript, Angular, NgRx

- Conducted **regression testing, integration testing, and performance validation** to ensure stability and accuracy of upgraded endpoints.
- Collaborated with **cross-functional teams (frontend, QA, DevOps)** to coordinate releases and minimize production defects.
- Utilized **Swagger and Postman** for API documentation, validation, and automated testing, ensuring clarity and alignment across engineering teams.
- Integrated frontend and backend components to ensure **secure communication, responsive interfaces, and seamless user experience across platforms**.
- Implemented **state management, reusable component architecture, and responsive UI design** to improve maintainability and scalability of the application.
- Conducted code reviews, implemented best practice in **Agile/Scrum** environments.
- Leveraged **Git, Azure Repos, and DevOps boards** for source control, task tracking, and project management.

Full Stack Software Development Engineer

American National Insurance

2020 – 2023

- Developed a Call Center Portal (CCP) application, architecting secure and scalable backend services with Node.js, **NestJS, TypeORM, MySQL**, and TypeScript, and building the frontend with JavaScript, **Vue.js**, and **Vuetify**.
- Implemented **OAuth 2.0** authentication and authorization flows, integrating role-based access controls and user security layers to ensure compliance and data protection.
- Designed and optimized backend services with **NestJS**, including API endpoints, middleware, and service modules to support business logic and microservices.
- Implemented **WebSocket-based real-time communication** to enable instant updates without page refreshes.
- Built **bi-directional data streams** between frontend and backend for live notifications, status updates, and events.
- Utilized Microsoft DevOps (Pipelines, Boards, Artifacts, Repos, Test Plans) to manage the full development lifecycle from planning through deployment.
- Designed, implemented, and tested Azure Serverless Functions in C# to support microservices and extend system capabilities.
- Developed services to consume and integrate external APIs, enabling dynamic search, filtering, and data management workflows.
- Delivered high-performance, scalable solutions by resolving complex data and design challenges.
- Led and contributed to multiple projects, enabling comprehensive data operations (GET, POST, DELETE) with secure access, modular design, and consistent integration patterns.

Full Stack Software Development Engineer

Improving Enterprise

2017 - 2019

- Managed application databases using SQL Server Management Studio (SSMS).
- Designed intuitive user interfaces with XAML to deliver an engaging user experience.
- Utilized Microsoft Azure for secure storage and management of mobile and web application data.

- Developed WPF applications with Microsoft technologies, aligning solutions with client specifications.
- Built robust, high-performance websites with a diverse technology stack, including React, JavaScript, Bootstrap 3, Vue.js, CSS, HTML, Angular 2+, ASP.NET, Entity Framework, and Fluent Validation.
- Implemented cross-platform and platform-specific features in C# using Xamarin, leveraging interfaces, Dependency Services, and Dependency Injection.
- Collaborated effectively using Git and Microsoft Team Foundation Server (TFS) for version control and team workflows.
- Integrated In-App Purchase functionality across App Store, Google Play Store, and Windows Store platforms.

Full Stack Software Developer

Hamper (Start Up)

2015 - 2017

- Designed, coded, enhanced and tested cross platform applications, from front-end to back-end.
- Build a team site using JavaScript, bootstrap 3, Vue.JS, CSS, HTML
- Developed cross-platform C# applications that allow customers to place and track orders
- Hamper is Uber for dry cleaning that runs a delivery dry cleaning service, using intelligent logistics and software to solve problems
- Used Selenium to run automation test on Hamper website